

Wireshark Practical for Windows

Practical 1: Installation of Wireshark

Objective

To install and configure Wireshark on Windows.

Steps

1. Download Wireshark from [Wireshark Official Website](https://www.wireshark.org)
2. Run the installer.
3. Install **Npcap** when prompted.
4. Complete the installation.
5. Launch Wireshark.

Result

Wireshark was successfully installed and ready for packet capturing.

Practical 2: Capturing Network Traffic

Objective

To capture packets on a network interface.

Steps

1. Open Wireshark.
2. Select the active network adapter (Wi-Fi/Ethernet).
3. Click the blue shark-fin icon (Start Capture).
4. Open a website in a browser.
5. Observe packets appearing in real time.
6. Click the red square icon to stop capturing.

Result

Network packets were successfully captured and displayed.

Practical 3: Filtering Packets

Objective

To analyze specific network traffic using filters.

Common Filters

Filter	Description
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS traffic
http	Show HTTP traffic
icmp	Show Ping packets
tcp.port == 80	Traffic on port 80
ip.addr == 192.168.1.1	Specific IP address

Steps

1. Start packet capture.
2. Enter a filter in the filter bar.
3. Press Enter.
4. Observe filtered packets.

Result

Only desired packets were displayed.

Practical 4: Ping Analysis

Objective

To capture and analyze ICMP packets.

Steps

1. Start Wireshark capture.
2. Open Command Prompt.
3. Execute:

ping google.com

4. In Wireshark apply:

icmp

5. Observe Echo Request and Echo Reply packets.

Result

ICMP communication between the host and destination was analyzed.

Practical 5: DNS Analysis

Objective

To observe DNS name resolution.

Steps

1. Start packet capture.
2. Open a website not recently visited.
3. Apply filter:

dns

4. Locate DNS Query and DNS Response packets.

Information Obtained

- Domain Name
- DNS Server IP
- Response Time
- Resolved IP Address

Result

DNS resolution process was successfully observed.

Practical 6: TCP Three-Way Handshake

Objective

To analyze TCP connection establishment.

Steps

1. Start capturing packets.
2. Visit a website.
3. Apply filter:

tcp

4. Locate packets with:
 - SYN
 - SYN-ACK
 - ACK

Handshake Process

Client → SYN

Server → SYN-ACK

Client → ACK

Result

TCP three-way handshake was successfully analyzed.

Practical 7: HTTP Packet Analysis

Objective

To analyze HTTP requests and responses.

Steps

1. Start packet capture.
2. Access an HTTP website.
3. Apply filter:

http

4. Observe:
 - GET requests
 - POST requests
 - HTTP responses

Result

HTTP communication was captured and analyzed.

Practical 8: FTP Traffic Analysis

Objective

To analyze FTP communication.

Filter

ftp

Observe

- Username
- Password (if unencrypted)

- Commands
- File transfers

Result

FTP packets were successfully analyzed.

Practical 9: Identifying Source and Destination IP Addresses

Objective

To determine communicating hosts.

Steps

1. Capture packets.
2. Select any packet.
3. Expand the Internet Protocol section.
4. Note:
 - Source IP
 - Destination IP

Result

IP addresses of communicating systems were identified.

Practical 10: Exporting Captured Packets

Objective

To save captured traffic for later analysis.

Steps

1. Capture packets.
2. Click **File** → **Save As**.
3. Save as:

capture.pcapng

4. Reopen the file when needed.

Result

Captured packets were successfully saved and reopened.

Practical 11: Find Your IP Address in Wireshark

Objective

Identify your system's IP address from captured packets.

Steps

1. Start packet capture.
2. Open Command Prompt.
3. Run:

ipconfig

4. Note your IPv4 address.
5. In Wireshark apply:

ip.addr == YOUR_IP

Result

Packets associated with your system IP were displayed.

Practical 12: ARP Packet Analysis

Objective

Observe Address Resolution Protocol (ARP).

Steps

1. Start capture.
2. Open Command Prompt.
3. Run:

```
arp -d *
```

```
ping google.com
```

4. Apply filter:

```
arp
```

Observe

- ARP Request
- ARP Reply
- MAC addresses

Result

ARP operation was successfully analyzed.

Practical 13: Analyze HTTPS Traffic

Objective

Capture encrypted web traffic.

Steps

1. Start capture.
2. Open:

```
https://www.google.com
```

3. Apply filter:

```
tls
```

Observe

- TLS Handshake
- Client Hello
- Server Hello

Result

Encrypted communication was observed.

Practical 14: Follow TCP Stream

Objective

View complete TCP conversation.

Steps

1. Capture traffic.
2. Select any TCP packet.
3. Right-click.
4. Click:

Follow → TCP Stream

Result

Entire client-server communication was displayed.

Practical 15: Detect Retransmissions

Objective

Identify retransmitted packets.

Filter

tcp.analysis.retransmission

Result

TCP retransmissions were identified.

Practical 16: Packet Coloring Rules

Objective

Understand packet color coding.

Steps

1. Capture traffic.
2. Observe different packet colors.
3. Open:

View → Coloring Rules

Result

Different protocols were identified using colors.

Practical No 17: Generate FTP Traffic

Open Command Prompt and run:

ftp ftp.gnu.org

Then in Wireshark use the filter:

ftp

If the connection succeeds, you'll see FTP commands and responses.